

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID : _____

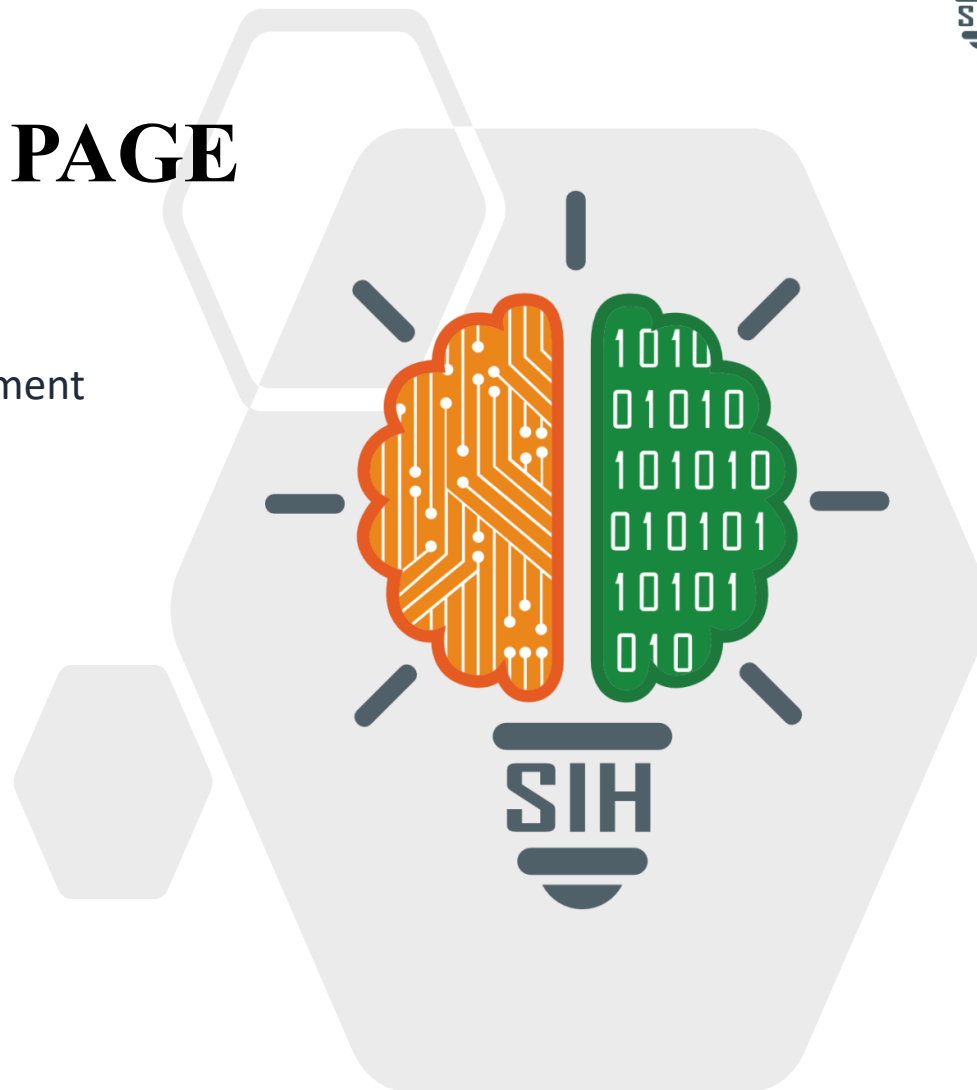
Problem Statement Title : Blockchain-Secured Digital Document
& Case Management

Theme : Smart Justice & Public Safety

PS Category : Software

Team ID : _____

Team Name (Registered on portal) : _____



Your
 Team
 Name

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

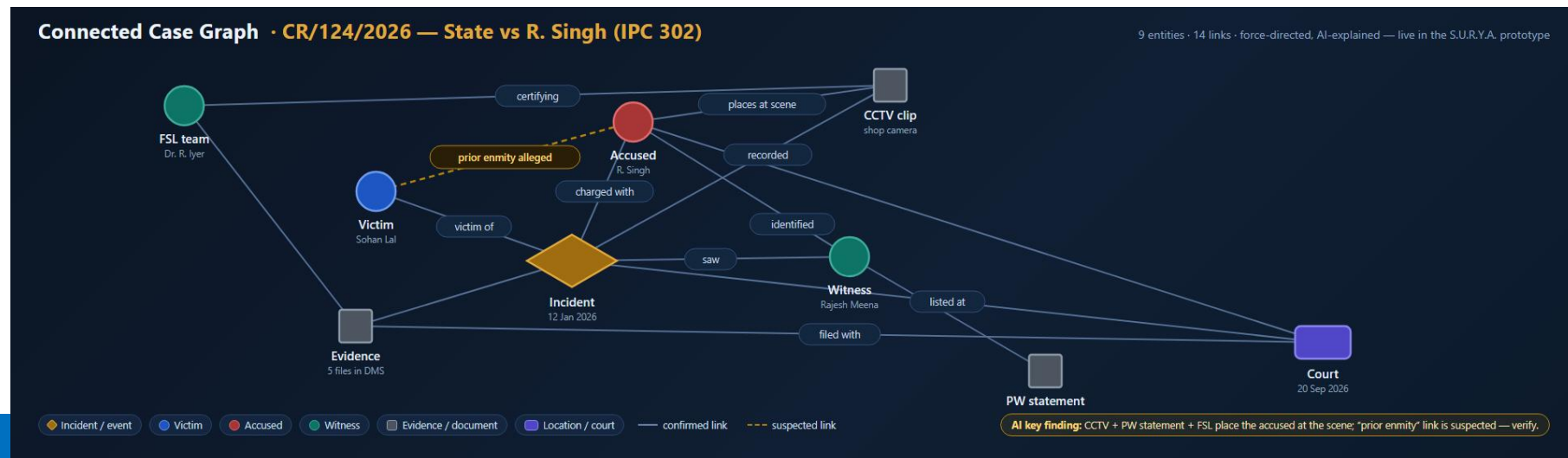
S.U.R.Y.A. (Smart Unified Resource for Judicial Assistance) is a blockchain-anchored digital Document Management System giving every case document — FIR, investigation record, witness statement, charge sheet, evidence record, forensic report, judgment — one tamper-evident, access-controlled, fully auditable digital life.

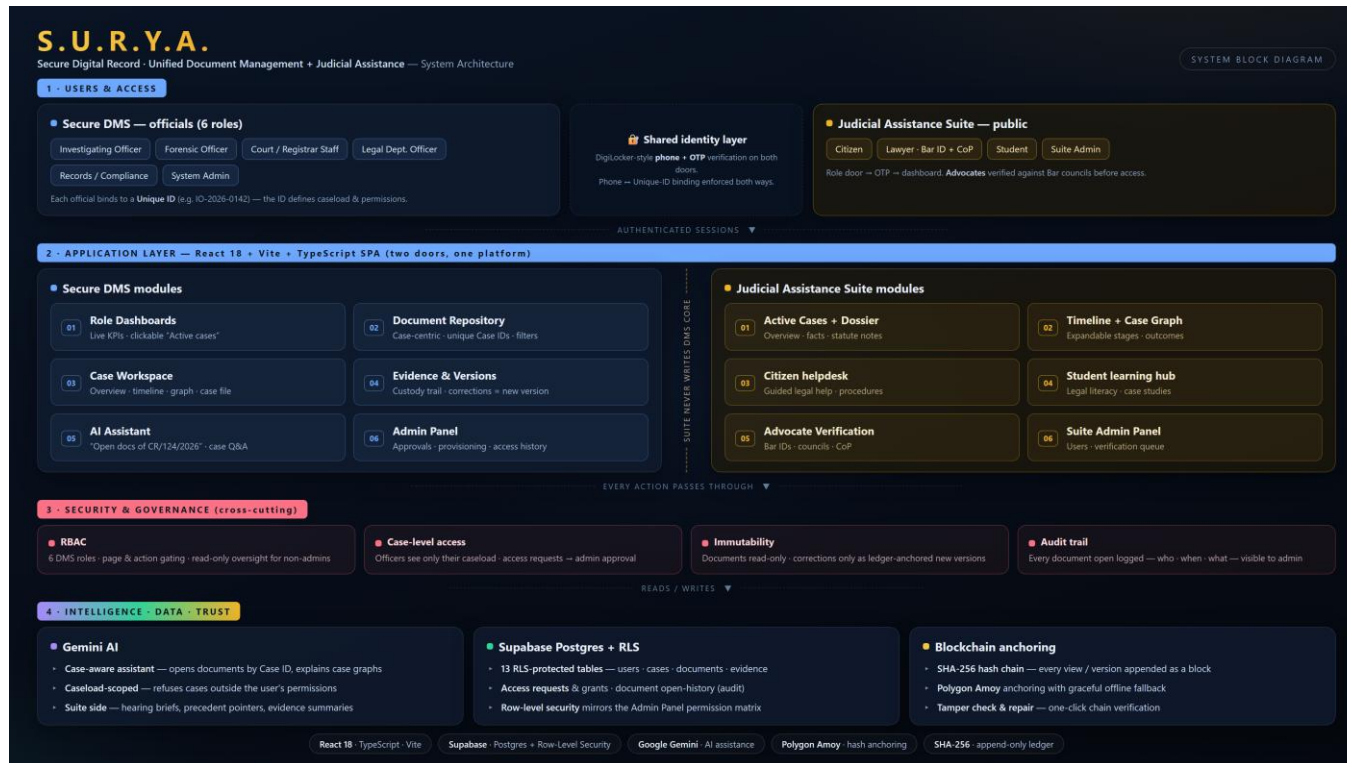
How it addresses the problem

- Unique Case ID (e.g. CR/124/2026) is the single key — retrieve all related documents instantly, ending scattered drives and email silos.
- Role-based access control (6 official roles) + case-level permission: cross-case needs go through access requests approved by the System Admin.
- SHA-256 hash chain + Polygon anchoring make tampering cryptographically detectable; corrections only as new ledger-anchored versions (version control).
- Every document open logged in an append-only audit trail; digital signatures & chain-of-custody for evidence records (IT Act § 65B-ready).

Innovation and uniqueness of the solution

- Trust layer, not storage layer: only hashes go on-chain — tampering becomes detectable, not just harder.
- Connected Case Graph (below, live in prototype): AI-explained network of victim, accused, witness, evidence & court — reveals suspected links like “prior enmity alleged” for verification, a first-of-its-kind bridge from a blockchain system of record to the integrated Judicial Assistance suite (citizen / lawyer / student).





Technologies to be used

React 18 + TypeScript (Vite SPA) · Supabase Postgres with Row-Level Security (13 RLS tables) · SHA-256 append-only hash ledger · Polygon Amoy anchoring (graceful offline fallback) · Google Gemini AI · DigiLocker-style Aadhaar-aligned OTP identity binding.

Methodology and process for implementation

- ① Phone + OTP identity verification; phone ↔ Unique-ID binding enforced both ways.
- ② Role-based dashboard loads the officer's active caseload (unique case IDs).
- ③ Case-centric repository: immutable viewer; every document open logged.
- ④ Upload/view/correction → SHA-256 hash → ledger block → Polygon Amoy anchor.
- ⑤ Cross-case need? Access request → System Admin approval → permissioned, logged access.
- ⑥ AI assistant retrieves permissioned documents and explains the case graph (left).

Architecture (block diagram, left)

Users & access → application layer → security & governance → intelligence · data · trust (Gemini AI, Postgres RLS, SHA-256 + Polygon anchoring).

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FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**

- Working prototype already demonstrates every claimed flow: OTP identity → caseload → case workspace → immutable documents → hash anchoring → access-request approvals → audit history.
- Built entirely on managed, government-deployable infrastructure (Supabase/Postgres, public blockchain testnet) — no exotic hardware; runs on commodity cloud; aligns with MeitY cloud-first posture.

Potential challenges and risks

- Legacy migration of paper records at scale · bandwidth-constrained PSUs · field-officer usability · identity spoofing · chain bloat · legal admissibility of e-records.

Strategies for overcoming these challenges

- Phased rollout (digitize-in-flight cases → back-scanning drives) with OCR-assisted bulk ingestion · offline-first PWA + low-bandwidth fallbacks · on-chain anchors only (tiny, cheap) with re-anchoring batching · IT Act 2000 § 65B-compliant certificates and audit exports for court admissibility.

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IMPACT AND BENEFITS



- **Potential impact on the target audience**

- Investigating officers & forensic labs: document retrieval in seconds by Case ID; zero ambiguity on “which version is current”; custody slips replaced by digital chain-of-custody.
- Courts & legal departments: verifiable, signed filings; disclosure workflows instead of email silos; faster hearings — less delay from document friction.
- Citizens, lawyers, students: the integrated S.U.R.Y.A. suite delivers legal aid, case learning and advocate verification on top of a trustworthy record layer.

Benefits of the solution (social, economic, environmental, etc.)

- Social: witness & minor identity protection through enforced confidentiality; public trust via independent tamper-evidence.
- Economic: paper, courier, storage and re-work savings across police/forensic/court back offices; leakage and litigation risk reduced.
- Environmental: millions of printed pages eliminated each year; Governance fit: RTI support, retention schedules, auditable compliance — ready for Ministry-grade infrastructure.

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RESEARCH AND REFERENCES



- **Details / Links of the reference and research work**

- IT Act 2000 & § 65B Evidence Act certification requirements — indiacode.nic.in
- National Data Sharing & Accessibility Policy; MeitY cloud-first policy — meity.gov.in
- eCourts Mission Mode Project (Phase III) & Virtual Courts — ecourts.gov.in
- Interoperable Criminal Justice System (ICJS) — mha.gov.in
- CCTNS — crime and criminal tracking network — mha.gov.in
- NIST SP 800-63 Digital Identity Guidelines; W3C Verifiable Credentials — nist.gov, w3.org
- Ethereum/Polygon official docs (proof-of-stability anchoring patterns) — polygon.technology
- DigiLocker API ecosystem (issuer/requester architecture) — digilocker.gov.in
- IEEE papers on blockchain-based document integrity & chain-of-custody (hash-chaining, Merkle anchoring).